

Friction and wear mechanisms of a thermoplastic composite GF/PA6 subjected to different thermal histories

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Abstract

An experimental investigation was conducted to identify the optimum manufacturing process that would improve the tribological properties of unidirectional GF/PA6 composite material. A range of microstructural morphologies were achieved by controlling the thermal history. Differential scanning calorimetry (DSC) was used to measure the crystallinity of the GF/PA6 samples subjected to different thermal histories. Friction tests were carried out on a scratch machine with a diamond indenter. To study the mechanisms of friction and wear, SEM and confocal laser microscope techniques were used. The results indicated that a slow cooling condition led to a higher percentage of crystallinity in the PA6 matrix, higher wear resistance and higher scratch coefficient. Greater wear was observed in many cases after a fast cool. It was also found that different levels of interfacial bonding between fibre and matrix played an important role.

Keywords: Composites; Crystallinity; Interface; Friction; Wear; Mechanisms

1. Introduction

Thermoplastic composite materials have replaced many classical materials like metals because of their high ratio of strength to weight, easy fabrication and outstanding chemical resistance. Glass fibres are the most widely used reinforcing agents for nylon. These fibres are usually sized to permit good bonding with the nylon matrix, producing a material of high flexural and tensile strength. Glass fibre-reinforced nylons have low water absorption, good dimensional stability, and low mould shrinkage [1]. They are being increasingly used in industry as bearing and sliding components in various environments. In the last decade the study of the tribological behaviour of these materials, such as the effects of fibre orientation on the coefficient of friction and the sliding velocity on the surface temperature and wear rate etc., has received much attention [2]. The microstructural changes and mechanical properties were improved in annealed PA6 samples studied by Russian scientists [3]. However, the relationship between microstructure of composites and tribological properties is not clear. In this paper GF/PA6 composite materials were selected to study the relationships between friction and wear mechanisms and the microstructure subjected to different thermal histories.

2. Specimen preparation

2.1. Materials

The material selected in this study was unidirectional GF/PA6 commingled yarn with a glass fibre volume fraction of 40%, supplied for testing by Toyobo Co. Ltd, Japan. The E-glass fibre diameter ranged from 10 to 15 μm , and the melting point T_m of the polyamide was 210 $^{\circ}\text{C}$ with a glass transition temperature T_g of 40 $^{\circ}\text{C}$. The composite density was 1.7 g cm^{-3} .

2.2. Manufacture of specimens

GF/PA6 commingled yarn was wound unidirectionally onto a steel plate at 40 rev min^{-1} , giving 15 yarns per in. The yarns were welded together along a line 10 mm away from the edge of the steel plate on both sides with a 40 W soldering iron. The GF/PA6 sheets were cut off from the steel plate along the edge. Fourteen commingled yarn sheets were laid up unidirectionally in a 200 mm square steel mould. The final thickness of the specimens was about 4.5 mm. The consolidation process was carried out on a hydraulic hot press (Moore G748), with a temperature of 245 $^{\circ}\text{C}$ and 1 MPa applied pressure on the top and bottom platens for 10 min.

To obtain the different microstructures in the composites, the moulds were cooled down at three different rates: (1)

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–1 °C min^{−1} in the hot-press; (2) –3 °C min^{−1} in air; and (3) –60 °C min^{−1} quenched in water. A K-type thermocouple, connected to a compensation box, was inserted into the mould and the electronic signal was collected by a computer at the speed of one data point per second to record the profile temperature during the whole consolidation process.

3. Testing procedure

3.1. Crystallinity measurement

Polyamide is a semi-crystalline thermoplastic, and there is always an amorphous phase existing in bulk composites [4]. To obtain information regarding the crystallinity of the GF/PA6 composites which were subjected to the three different cooling conditions, differential scanning calorimetry (DSC) measurements were performed. The test was carried out on a Mettler TA 4000 system with a small sample (about 10 mg) cut off from the bulk specimen.

For the slowest cooling condition sample, the melting point is taken at the peak temperature, 226 °C, where melting is virtually complete. The peak area is equal to H_m , which is the heat of fusion or the fusion endotherm of the crystal structure of the PA6 in the sample.

To calculate the percentage of crystallinity in the PA6, one must know exactly how much PA6 is in the GF/PA6 composite of the DSC sample. So after DSC analysis, the sample with a weight W_c was heated up to 500 °C in an oven for 6 h, burning the PA6 away, then the glass fibre was weighed on an analytical balance (sensitivity 0.01 mg) in the same atmosphere (W_G).

The percentage of the crystallinity of the PA6 was calculated from the following expression:

$$\% \text{ crystallinity} = (\Delta H_m / \Delta H_s) \times 100$$

where ΔH_s is the specific fusion endotherm of 1 g PA6 fully crystallized, 190 J g^{−1} [5], and ΔH_m is the measured fusion endotherm of 1 g PA6 in the sample calculated from:

$$\Delta H_m = H_m / (W_C - W_G)$$

3.2. Transverse flexural test

A three-point bending test was carried out on an Instron 4302 to study the influence of thermal history on the GF/PA6 composites' interfacial bonding condition. The specimen was loaded along the fibre orientation. The test method followed the ASTM standard D790M-84a. The transverse flexural strength was calculated from:

$$\sigma_b = \frac{3PL}{2bd^2}$$

where P is the fracture load (N), L is the support span (mm), b is the width (mm), and d is the thickness of beam tested (mm).

3.3. Scratching test

Dry single pass scratching tests were performed at room temperature with a pyramidal diamond indenter with a tip angle of 136° and a constant normal load of 10 N to study the influence of thermal history on the friction and wear mechanisms of GF/PA6 composites. The orientation of the pyramidal indenter was such that one leading plane moved ahead during the scratching test and the sliding speed was kept constant at 6 mm s^{−1}. The specimens measuring 10 × 10 × 4.5 mm³ were polished and fixed to a movable steel base with wax. Three sliding directions of five specimens were tested: (a) along the fibre direction (parallel direction); (b) across the fibre direction (transverse direction); and (c) across the ends of the fibre (normal direction).

The tangential force was measured by a strain gauge transducer and the average value for each scratch was determined from 15 data points mid-length of the scratch.

3.4. Microscopic study

The scratching and transverse flexural specimens were examined after testing using a Confocal Microscope, Laser MRC 600 3-D reconstruction, and a Philips 505 Scanning Electronic Microscope. The specimens were sputter coated with a thin layer of gold in order to improve the resolution of the specimen prior to SEM examination. For the low resolution of GF/PA6 an accelerating voltage was selected at 20 kV and the condenser lens (spot size) was selected as 100 nm, both backscatter and secondary electron detectors were used. Polarized light was used to examine thin films of the glass fibre in PA6, prepared in the same way as the bulk specimens.

4. Results and discussion

4.1. Crystallinity

The percentage of crystallinity in the PA6 matrix of the unidirectional composites sample subjected to three different thermal histories was evaluated from DSC analysis. The results indicated that the PA6 matrix crystallinity in the GF/PA6 composites was dependent on the thermal history. There were clear differences between the cooling conditions 1, 2 and 3 (see Fig. 1). Cooling in a hot press resulted in the highest crystallinity (X) at about 37%, followed by cooling in air of 33% and quenching in water, crystallinity of 28%. In general, an amorphous polymer has a random distribution of chains in its main molecular structure and this gives rise to the viscoelastic nature of the PA6. A crystalline polymer shows considerable three-dimensional order and there are dramatic changes in such properties as density, elasticity, plasticity and hardness. Therefore, there are differences not only in mechanical responses, but also in tribological response as compared with the amorphous variety.

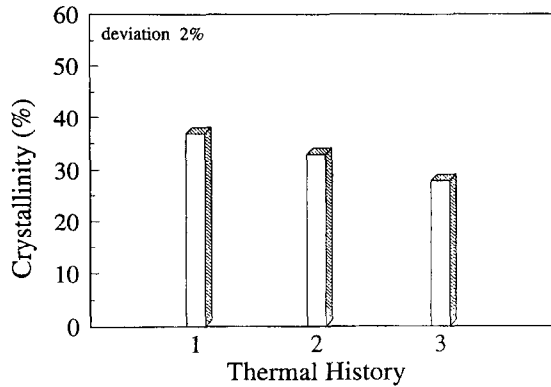


Fig. 1. The effect of thermal history on crystallinity of PA6, cooling rate of sample: (1) $-1^{\circ}\text{C min}^{-1}$; (2) $-3^{\circ}\text{C min}^{-1}$; (3) $-60^{\circ}\text{C min}^{-1}$.

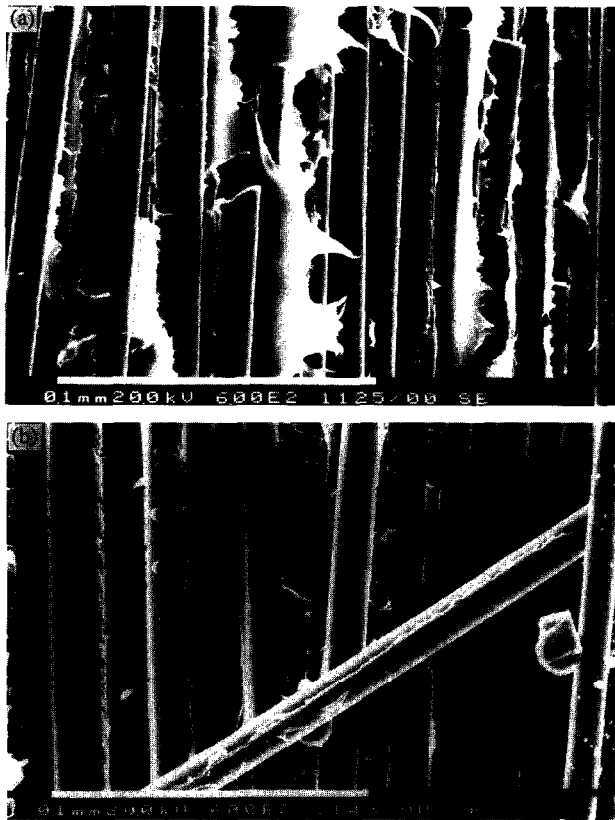


Fig. 2. SEM photo of transverse flexural fracture surface: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

4.2. Transverse flexural strength

SEM observations show that the rupture of the flexural specimens occurred between the fibre and the matrix (see Fig. 2(a) and (b)). This means the flexural strength could, in an indirect way, relate to the interfacial bond strength. Thus it can be inferred that the interfacial strength is largest for the slowest cooling rate, as shown in Fig. 3. It was also found that there are matrix ductile tears in slow cooling samples and matrix brittle rupture in fast cooling samples. Consequently, the crystallinity of the matrix contributes to the flexural strength as well.

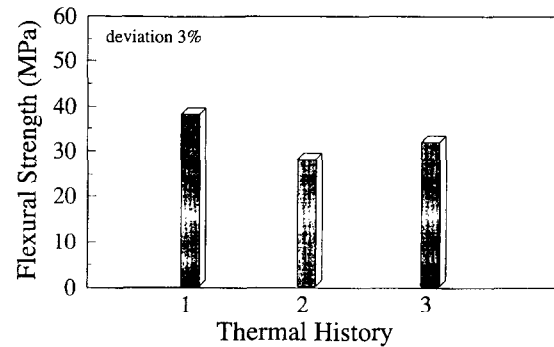


Fig. 3. Transverse flexural strength vs thermal history, cooling rate of sample: (1) $-1^{\circ}\text{C min}^{-1}$; (2) $-3^{\circ}\text{C min}^{-1}$; (3) $-60^{\circ}\text{C min}^{-1}$.

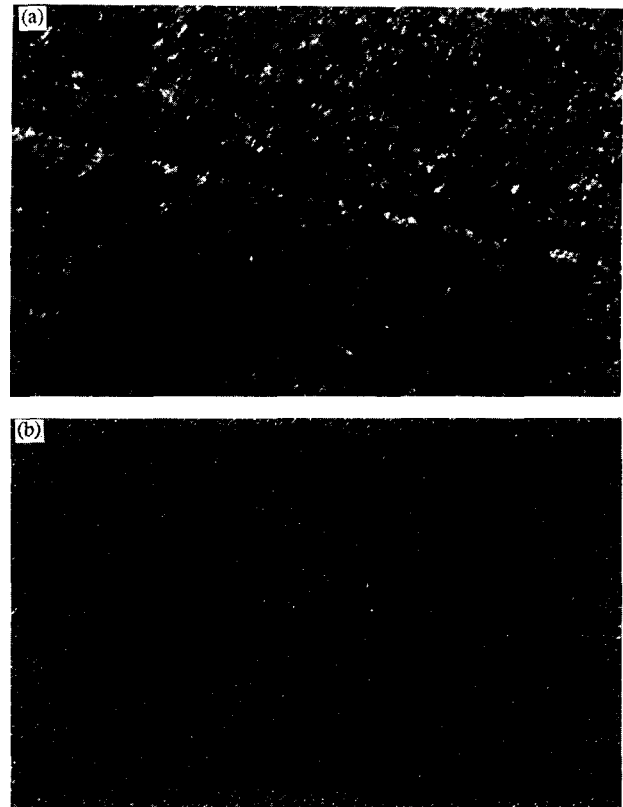


Fig. 4. Transmitted polarized light images of GF/PA6 thin film: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

The photographs of transmitted polarized light image of GF/PA6 thin film, Fig. 4, show a different crystalline layer between the fibre and the matrix in the two types of sample cooling conditions, which may be the reason for an improved interfacial bond and/or higher value of transverse flexural strength. The magnification of the images is $\times 200$.

4.3. Friction mechanisms

From SEM observations, Figs. 5–7, it can be seen that the main mechanisms of friction are fibre fracture, matrix ductile flow in slow cooling samples, brittle rupture in fast cooling samples, and interfacial debonding.

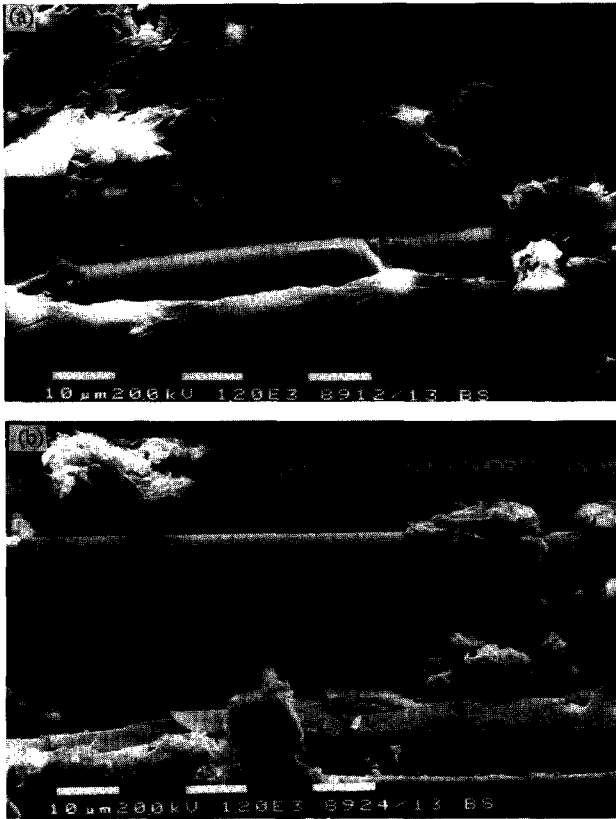


Fig. 5. SEM photos of parallel scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

In this paper, we determine the force ratio as a scratching coefficient μ_s , i.e.

$$\mu_s = \frac{F_t}{N}$$

where F_t is the average value of tangential force and N is the normal load applied on the diamond indenter. Bowden and Tabor [6] postulated that the friction force is composed of the adhesion and plowing terms for single phase metals. Zhang et al. [7] suggested that the friction force in a metal matrix composites is composed of three terms: plowing, adhesion and fracturing. From the observation of the GF/PA6 scratches with optical and electronic microscopes, the tangential force of the present composite can be considered to consist of the following four important components:

$$F_t = F_f + F_a + F_i + F_p$$

(1) F_f fibre fracture force; (2) F_a adhesion force; (3) F_i interfacial debonding; and (4) F_p plowing force. In the same sliding direction but different cooling conditions, the friction caused by breaking the fibres, plowing the matrix, adhesion and interfacial debonding terms is different due to different crystallinity and interfacial bonding conditions. Fig. 8 shows the scratching coefficient in the parallel, transverse and normal directions with different thermal history. The results indicate that slower cooling rate gives a higher value of scratching coefficient in all fibre orientations/scratching directions.

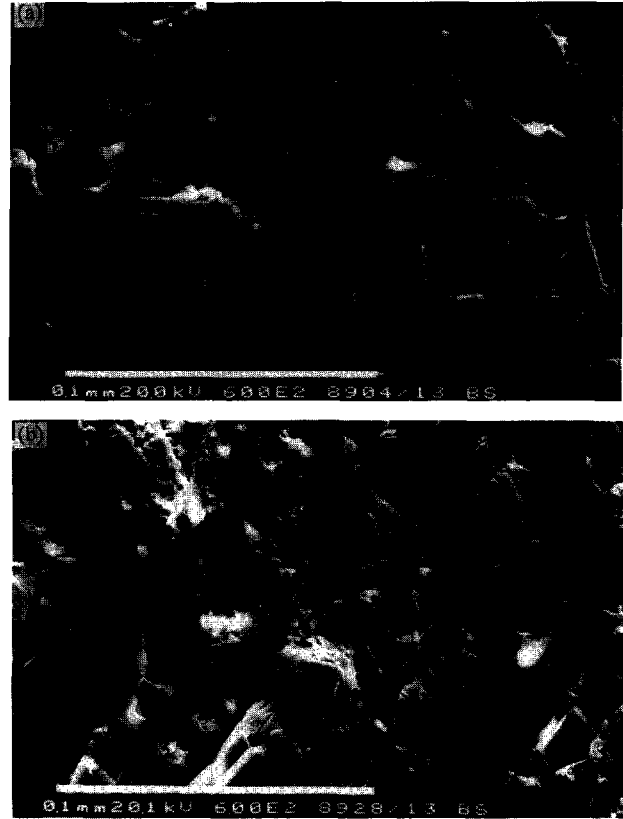


Fig. 6. SEM photos of transverse scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

In the parallel direction, as shown in Fig. 5, the dominant failure mechanisms for slow cooling samples is matrix ductile flowing, fibre–matrix debonding, and fibre fracturing into short fragments. However, on the fast cooling sample the matrix is brittle, rupturing into small debris, fibre–matrix debonding and the fibre fragments are longer than the slow cooling samples. Consequently, the fibre fracture term is greater in slow cooling samples than the fast one. The difference is slight in the adhesion force term between different cooling rate samples. Good interfacial bonding will resist the matrix peeling off from the fibre and stress will transfer over shorter lengths of fibre. Thus, the interfacial debonding force is higher in the slow cooling samples than the fast ones, as shown previously with the flexural test. It is also found that fibre and matrix were plowed out by the indenter under the shear stress. The cohesion is greater in the crystalline matrix than in the amorphous matrix so the plowing force is higher in the higher crystalline matrix which is the slow cooled sample, as compared to the lower crystalline matrix from the fast cooled sample.

In the transverse direction, the fibres were hooked out by the indenter and matrix chipping was observed in the fast cooling sample (see Fig. 6). The scratching coefficient is highly dependent on the interfacial bonding conditions. For better interfacial bond samples the materials' ultimate strength is higher and hence, to break greater interfacial bond, samples need higher stress. The three terms, F_f , F_a and F_p ,

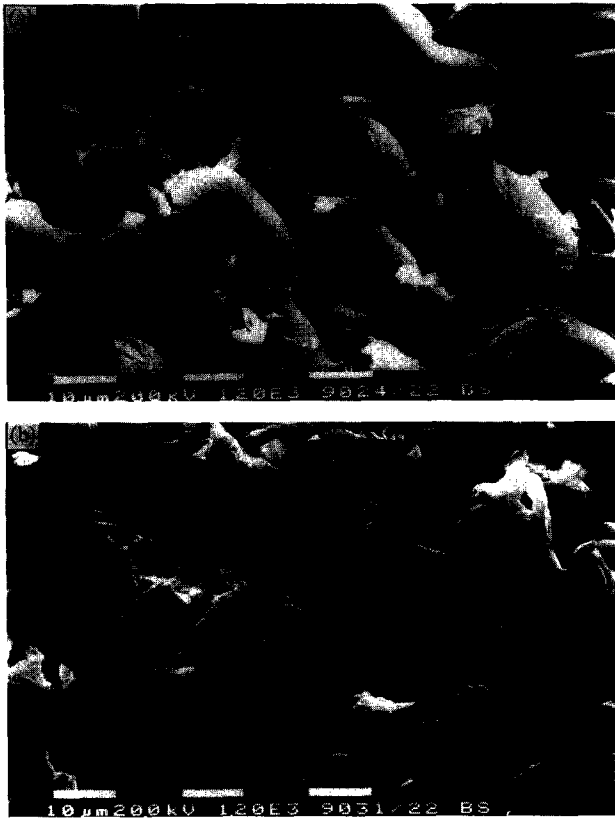


Fig. 7. SEM photos of normal scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

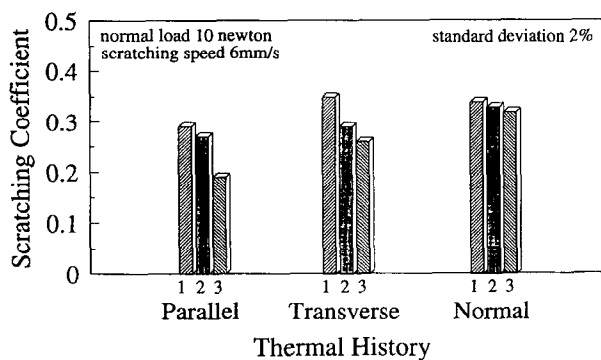


Fig. 8. Scratching coefficient vs thermal history, cooling rate of sample: (1) $-1^{\circ}\text{C min}^{-1}$; (2) $-3^{\circ}\text{C min}^{-1}$; (3) $-60^{\circ}\text{C min}^{-1}$.

were influenced by cooling rate similar to the parallel direction, so that the scratching coefficient is higher in the slow cooling samples.

In the normal direction, the dominant failure mechanisms in the slow cooling sample are the fibre pulling-out, interfacial debonding, and matrix ductile deformation. However, fibre ends fracture, interfacial debonding, and matrix brittle rupture are dominant mechanisms in the fast cooling sample (Fig. 7). The scratching coefficient varies slightly with the cooling rate in the normal scratching direction. The failure mechanisms are complicated and further study must be done.

The values of scratching coefficient in the parallel direction are the lowest for all cooling conditions. This is because the

failure mechanisms in the parallel direction do not involve as much fibre fracture as those in the transverse and normal scratches. With the same cooling rate but different scratching directions, the difference in fibre fracturing force, F_f , between the three scratching directions (P , T and N) is greater than the terms F_a , F_i and F_p .

4.4. Wear mechanisms

According to the above analysis, it is easier to plow the fast cooling sample than the slow cooling one by the indenter. It can be inferred that the wear resistance is higher in the slow cooling sample than the fast one. Indeed this can be seen from the two SEM photos of scratches on parallel direction samples, Fig. 9, which show obvious differences in the scratching wear mechanisms between the slow cooling rate sample 1 ($-1^{\circ}\text{C min}^{-1}$) and the fast cooling rate sample 3 ($-60^{\circ}\text{C min}^{-1}$). There is more matrix debris formed in the scratch groove of the fast cooling sample than that in the slow cooling sample. This is thought to be due to the larger amorphous content of the fast cooling samples and weak interfacial bonding. For abrasive wear, increasing crystallinity of pure polymers can cause the wear to decrease [8]. Generally, amorphous morphology is softer and more viscoelastic than a crystalline morphology in polymeric materials. Thus the cohesive energy or ultimate strength is also lower in the amorphous region. Consequently, the maximum stress

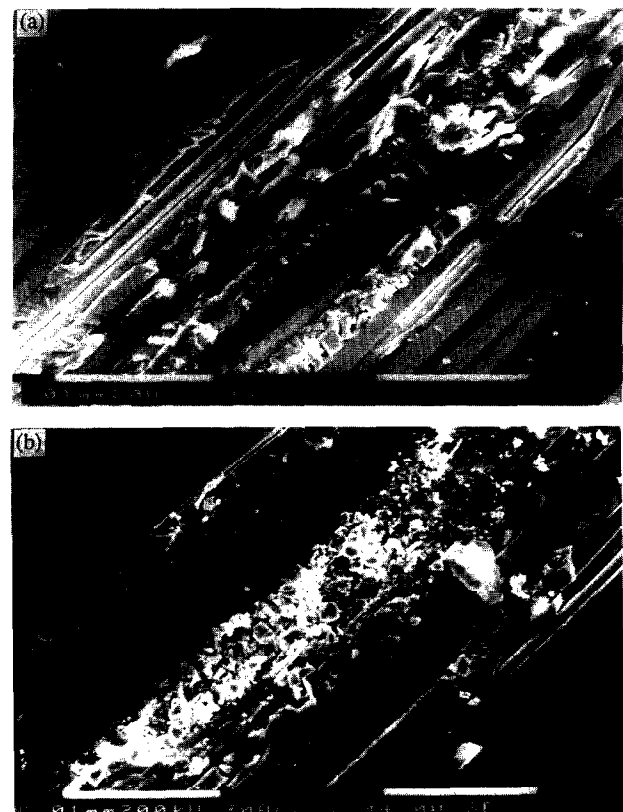


Fig. 9. SEM photos of parallel scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

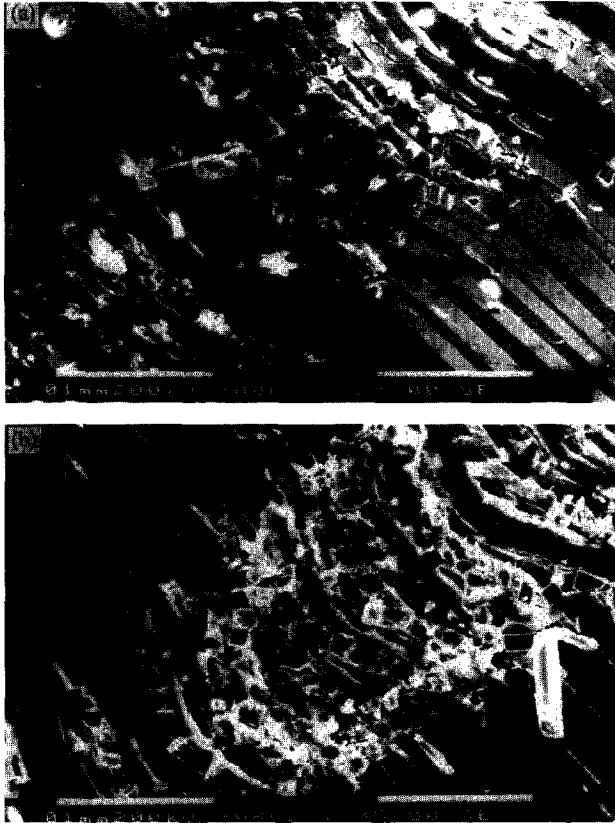


Fig. 10. SEM photos of transverse scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

exceeds the ultimate strength of the materials, and cracks are initiated and propagated until wear debris are produced. From the SEM photos, Fig. 9, which are the parallel direction scratches, it can be seen that the glass fibre and the PA6 matrix were ploughed out by the indenter.

Comparing the scratching grooves of the slow and fast cooling samples using laser confocal microscopy, it is found that the form has a wider and deeper groove in the fast cooling sample than in the slow one. That can imply a lower wear resistance in the fast cooling sample. It can be seen by the confocal microscope the groove shape on the sample does not match the shape of the indenter. This may be caused by the elastic recovery of the PA6 matrix material after the scratching.

In the transverse scratching test the mechanisms of wear are very complex. From the SEM photos (Fig. 10) it can be seen that a large number of fibres are debonded and fractured by the indenter. The matrix was plastically deformed, displaced and fractured under the shearing stress. It is the same for the parallel direction, with more debris formed in the scratch groove of the fast cooling sample than that in the slow sample.

Scratching in the normal direction shows micro-cracking and plastic deformation in the matrix of the slow cooling sample. Wear debris comes from both the fibre and matrix. There is also interfacial debonding between the PA6 matrix and the glass fibre. In the fast cooling condition there is a

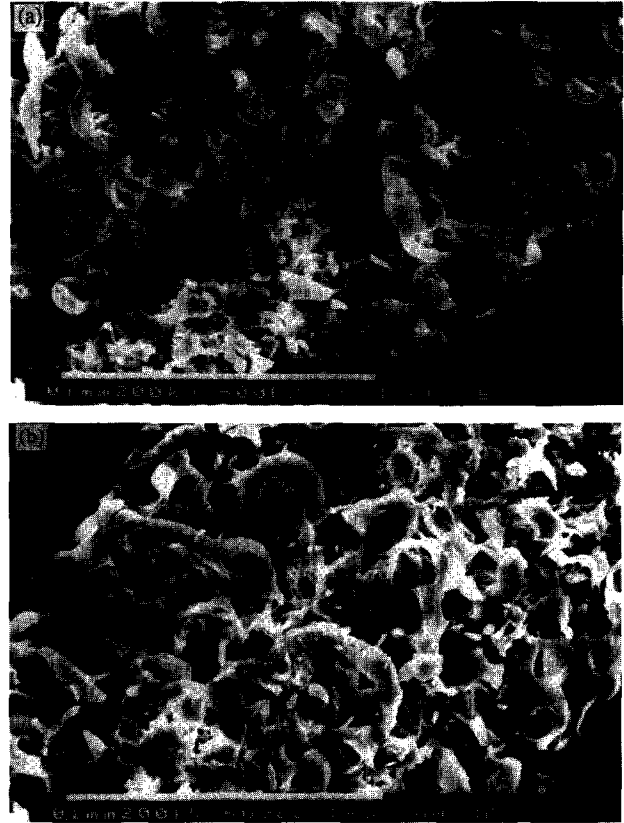


Fig. 11. SEM photos of normal scratches: (a) cooling condition 1 sample; (b) cooling condition 3 sample.

large amount of interfacial debonding and matrix cracking (see Fig. 11). It may be considered that a weak interfacial bond and an amorphous matrix would result in more severe damage due to abrasive wear. Generally, abrasive wear is inversely proportional to the hardness of the softer counterpart. High crystallinity gives a hard matrix in the GF/PA6 composite and strong interfacial bonding, thus limiting microcracking.

5. Conclusions

A slow cooling condition leads to a high crystallinity in PA6 matrix, which may result in a harder matrix and improves the interfacial properties. The friction and wear mechanisms are very different with a change in the cooling rate and this seems to be associated with the crystallinity of the matrix as well as the interfacial bonding strength. The scratching coefficient and wear resistance are higher in the slower cooling samples.

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